## 3.3 long run costs

**1.** A farmer grows wheat using two inputs: labor and land whose prices are constant. If she doubles her inputs, she finds that the quantity of wheat produced more than doubles. Therefore, it must be true that in this output range her long-run average total cost curve is

(A) upward sloping
(B) downward sloping
(C) horizontal
(D) vertical
(E) U-shaped

**2.** Which of the following MUST be true of the long run?

(A) It is at least one year in duration.
(B) All factors of production are variable.
(C) At least one factor of production is fixed.
(D) Marginal costs are constant.
(E) Average total costs are constant.

**3.** A merger of two firms may increase economic efficiency by

(A) decreasing average total cost through an increase in economies of scale
(B) decreasing output to reduce marginal cost and equalize price
(C) increasing economic profits but decreasing consumer surplus
(D) increasing consumer surplus by decreasing economic profits
(E) increasing consumer surplus by shifting the demand curve for the product to the right

**4.** Suppose a firm's production process exhibits diseconomies of scale. How and why will costs change if the firm reduces its output?

(A) Long-run average total cost will increase because it becomes more difficult for the firm to manage its workforce.
(B) Long-run average fixed costs will increase because the firm produces fewer units of output.
(C) Long-run average total cost will decrease because it becomes easier for the firm to manage its workforce.
(D) Long-run average fixed costs will decrease because the firm produces fewer units of output.
(E) Long-run average total cost will decrease because it becomes more difficult for the firm to manage its workforce.

**5.** What does the minimum efficient scale measure?

(A) The smallest output level at which short-run average total cost is minimized.
(B) The smallest output level at which short-run total cost is minimized.
(C) The smallest output level at which long-run average total cost is minimized.
(D) The largest output level at which long-run average total cost is minimized.
(E) The largest output level at which long-run total cost is minimized.

## 3.3 long run costs

**6.**

UNITS OF OUTPUT

| Units of Labor \ Units of Capital | 1 | 2 | 3 |
|---|---|---|---|
| 1 | 100 | 140 | 160 |
| 2 | 140 | 200 | 240 |
| 3 | 160 | 240 | 300 |

The table above shows the various units of output that can be produced with different combinations of capital and labor. Which of the following statements is correct according to the information in the table?

(A) In the long run, there are constant returns to scale.

(B) In the long run, there are increasing returns to scale.

(C) In the short run, the marginal product of capital is constant.

(D) In the short run, the marginal product of labor is constant.

(E) In the short run, the law of diminishing marginal returns does not hold.

**7.** If the output of a firm doubles when the firm doubles all of its inputs, the firm must be experiencing

(A) economies of scale

(B) increasing returns to scale

(C) constant returns to scale

(D) decreasing returns to scale

(E) diseconomies of scale

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The table below shows the long-run total cost function of a firm.

| Quantity of Output | Total Cost ($) |
|---|---|
| 0 | 0 |
| 1 | 10 |
| 2 | 20 |
| 3 | 30 |
| 4 | 40 |
| 5 | 50 |

**8.** The firm's cost function exhibits

## 3.3 long run costs

(A) diseconomies of scale
(B) constant returns to scale
(C) economies of scale
(D) decreasing marginal cost
(E) increasing marginal cost

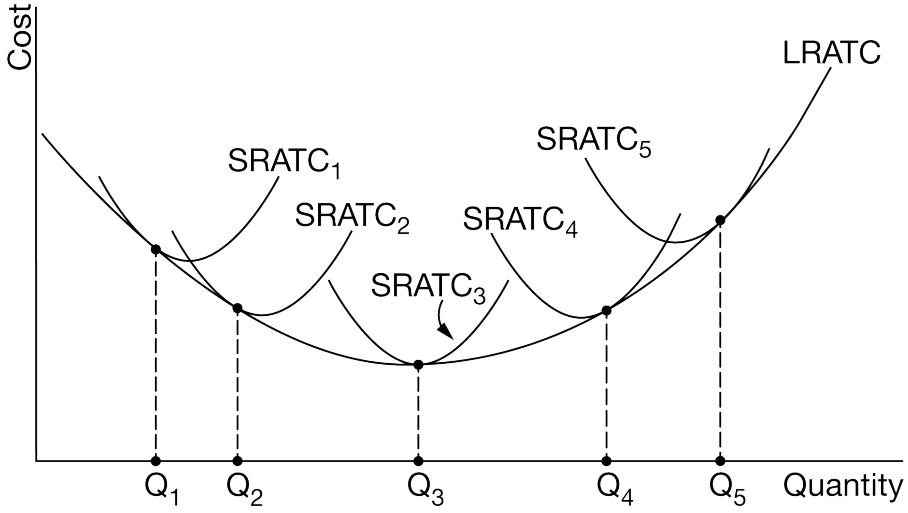
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**9.** Economies of scale can be illustrated by

(A) an increasing short-run marginal cost curve as a firm produces more output
(B) a decreasing short-run average total cost curve as a firm produces more output
(C) a downward-sloping long-run supply curve for an industry
(D) a decreasing long-run average total cost curve as a firm produces more output
(E) an increasing long-run average total cost curve as a firm produces more output

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A firm has only five possible factory (plant) sizes to choose from, represented by the short-run average total cost $(\mathrm{SRATC})$ curves on the long-run average total cost $(\mathrm{LRATC})$ curve shown on the graph below.




**10.** Which of the following ranges of output illustrates diseconomies of scale?

(A) $Q_1$ to $Q_2$
(B) $Q_1$ to $Q_3$
(C) $Q_2$ to $Q_3$
(D) $Q_2$ to $Q_4$
(E) $Q_3$ to $Q_5$

**11.** The firm's minimum efficient scale occurs on

## 3.3 long run costs

(A) $\mathrm{SRATC}_1$
(B) $\mathrm{SRATC}_2$
(C) $\mathrm{SRATC}_3$
(D) $\mathrm{SRATC}_4$
(E) $\mathrm{SRATC}_5$

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**12.** Which of the following is a result of increasing returns to scale?

(A) Upward-sloping short-run marginal cost curve
(B) Downward-sloping marginal physical product of labor curve
(C) Downward-sloping long-run average total cost curve
(D) Diseconomies of scale
(E) Diminishing returns

**13.** As a firm expands its output, it initially experiences economies of scale and then diseconomies of scale. The firm's long-run average cost curve is

(A) horizontal and then upward sloping
(B) horizontal and then downward sloping
(C) downward sloping and then upward sloping
(D) continuously upward sloping
(E) continuously downward sloping

**14.** Which of the following indicates the presence of economies of scale as the quantity of output increases?

(A) Average variable cost decreases.
(B) Long-run average total cost decreases.
(C) Marginal cost decreases.
(D) Average fixed cost decreases.
(E) Marginal cost exceeds average total cost.

**15.** When a firm is experiencing economies of scale, an increase in the firm's output will result in which of the following?

(A) A decrease in total cost
(B) A decrease in long-run average total cost
(C) A downward shift in the long-run marginal cost curve
(D) An increase in long-run average variable cost
(E) An increase in average fixed cost

**16.** If a firm experiences economies of scale over the entire range of output, the long-run average cost curve will be

(A) U-shaped
(B) upward sloping
(C) downward sloping
(D) horizontal
(E) below the marginal cost curve

---

## 3.3 long run costs

**17.** Which of the following is true about economies of scale and increasing returns to scale?

(A) Economies of scale and increasing returns to scale are the same thing.

(B) Economies of scale refers to the relationship between inputs and output. Increasing returns to scale refers to the relationship between long-run average total cost and the size of the firm.

(C) Economies of scale refers to the relationship between long-run average total cost and the size of the firm. Increasing returns to scale refers to the relationship between inputs and output.

(D) Economies of scale is a long-run concept, while increasing returns to scale is a short-run concept.

(E) Increasing returns to scale is a long-run concept, while economies of scale is a short-run concept.

**18.** Economies of scale exist when

(A) the doubling of all inputs doubles the output produced

(B) short-run average total cost decreases as output increases

(C) short-run average total cost remains constant as output increases

(D) long-run average total cost increases as output increases

(E) long-run average total cost decreases as output increases

**19.** If a firm is experiencing economies of scale, which of the following will decrease as output increases?

(A) Fixed cost

(B) Long-run total cost

(C) Long-run average total cost

(D) Marginal cost

(E) Marginal revenue

**20.** If a firm's production process exhibits economies of scale, which of the following will occur when the firm's output increases?

(A) Its short-run average total costs will rise.

(B) Its long-run average total costs will rise.

(C) Its long-run average total costs will fall.

(D) Its short-run average total costs will fall.

(E) Its long-run total costs will fall.

**21.** Which of the following must be true if a firm is experiencing economies of scale?

(A) All costs are explicit.

(B) Long-run average total cost decreases as the firm’s output increases.

(C) Economic profits decrease as the firm’s output increases.

(D) Long-run average total cost remains constant as the firm’s output decreases.

(E) Proportionate increases in inputs result in less-than-proportionate increases in output.

**22.** F&D Manufacturing Company increases all its inputs by 50 percent each. If F&D’s output increases by 100 percent, then F&D is experiencing

## 3.3 long run costs

(A) increasing returns to scale
(B) constant returns to scale
(C) diseconomies of scale
(D) increasing marginal cost
(E) decreasing profits

**23.** Gale Furniture uses labor, capital, and wood as inputs in its production of office desks. When Gale Furniture doubles the quantity of labor, capital, and wood, the quantity of office desks doubles. Gale Furniture's production process is characterized by

(A) economies of scale
(B) diseconomies of scale
(C) constant returns to scale
(D) decreasing returns to scale
(E) increasing returns to scale

**24.** Currently, XYZ Corporation can produce 50 units of output using 20 workers and 8 units of capital. Which of the following changes in the number of workers, units of capital, and quantity of output are consistent with constant returns to scale?

(A)

| Workers | Capital | Output |
|---|---|---|
| 40 | 8 | 100 |

(B)

| Workers | Capital | Output |
|---|---|---|
| 40 | 16 | 90 |

(C)

| Workers | Capital | Output |
|---|---|---|
| 20 | 4 | 25 |

(D)

| Workers | Capital | Output |
|---|---|---|
| 10 | 4 | 25 |

(E)

| Workers | Capital | Output |
|---|---|---|
| 10 | 8 | 25 |

## 3.3 long run costs

**25.** If a firm's long-run average total cost increases as output increases, the firm is experiencing

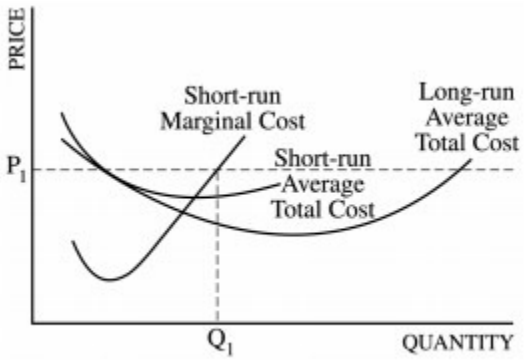
(A) economies of scale
(B) diseconomies of scale
(C) increasing returns to scale
(D) efficiency in plant size
(E) maximum economic profit

**26.** If a single firm can produce and supply an entire market at a lower unit cost than many small firms can, the long-run average total cost must be

(A) increasing as firm size increases
(B) remaining constant as firm size increases
(C) decreasing as the firm's output increases
(D) inelastic due to specialization
(E) constant and equal to marginal cost

---

The following questions refer to the graph below, which shows the cost curves of a firm.




**27.** If the firm produces $Q_1$ units of output with two inputs, the firm will be experiencing which of the following in the short run and in the long run?

## 3.3 long run costs

(A)

| Short Run | Long Run |
|---|---|
| Increasing marginal returns | Economies of scale |

(B)

| Short Run | Long Run |
|---|---|
| Increasing marginal returns | Diseconomies of scale |

(C)

| Short Run | Long Run |
|---|---|
| Diminishing marginal returns | Economies of scale |

(D)

| Short Run | Long Run |
|---|---|
| Diminishing marginal returns | Diseconomies of scale |

(E)

| Short Run | Long Run |
|---|---|
| Constant marginal returns | Diseconomies of scale |

---

**28.** In most cases the supply curve for a perfectly competitive industry can be described as which of the following?

(A) More elastic in the short run than in the long run
(B) More elastic in the long run than in the short run
(C) Downward sloping in the short run
(D) Perfectly inelastic in the long run
(E) Perfectly elastic in the short run

**29.** Under which of the following circumstances is a firm experiencing economics of scale?

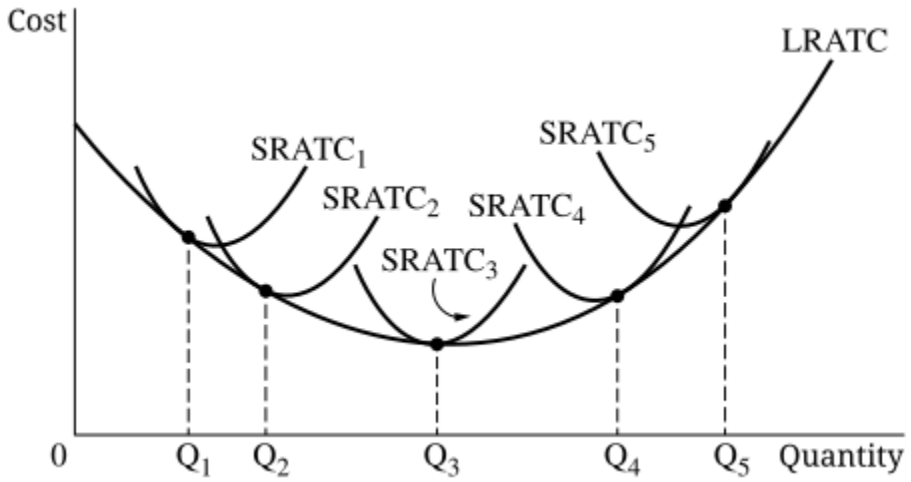
(A) The firm increases only its labor input, and output decreases.
(B) The firm doubles its inputs, and output triples.
(C) The firm builds a new plant, and the average cost of production increases.
(D) The firm hires a new plant manager, and profits increase.
(E) The product price increases, and the firm increases its output.

**30.** The long-run average cost curve will be sloping downward if a firm experiences

## 3.3 long run costs

(A) diminishing marginal returns
(B) decreasing returns to scale
(C) constant returns to scale
(D) diseconomies of scale
(E) economies of scale

**31.**




The graph provided shows five different short-run average total cost ($\mathrm{SRATC}$) curves of a firm and its long-run average total cost ($\mathrm{LRATC}$) curve. Which of the following is true if the firm is operating on its $\mathrm{SRATC}_2$ curve and producing quantity $\mathrm{Q}_2$?

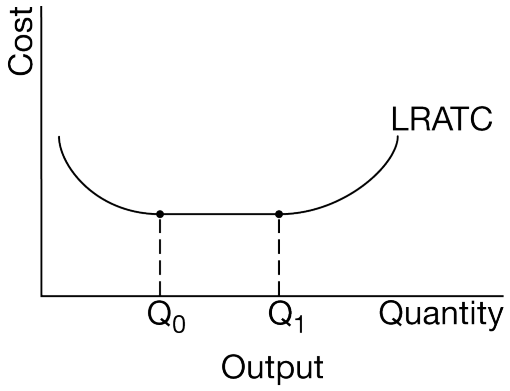
(A) The firm is producing at the minimum efficient scale.
(B) The firm must decrease its output to achieve the minimum efficient scale.
(C) The firm must increase its output to achieve the minimum efficient scale.
(D) The firm is producing at its productively efficient quantity of output.
(E) The firm is experiencing diseconomies of scale.

**32.** Which of the following indicates that a firm is experiencing economies of scale?

(A) The firm's long-run supply curve is horizontal.
(B) The firm's long-run marginal cost increases as output increases.
(C) The firm's long-run average total cost decreases as output increases.
(D) The firm's long-run total cost decreases as output increases.
(E) The firm's total revenues increase as output sold increases.

## 3.3 long run costs

**33.**




The graph above shows a firm's long-run average total cost curve $(\mathrm{LRATC})$. Which of the following statements is true as the firm increases its scale of production?

(A) From zero output to $Q_0$, the firm experiences diseconomies of scale.

(B) From $Q_0$ to $Q_1$, the firm experiences increasing returns to scale.

(C) For output levels above $Q_1$, the firm experiences diseconomies of scale.

(D) At all output levels, the firm's long-run marginal cost must be constant.

(E) From $Q_0$ to $Q_1$, $\mathrm{LRATC}$ falls because average fixed cost decreases; after $Q_1$, $\mathrm{LRATC}$ rises due to diminishing returns to labor.

**34.** Assume a firm doubles its usage of each input, resulting in a doubling of the firm's output. Which of the following describes this result?

(A) Increasing total cost

(B) Diminishing marginal returns

(C) Decreasing returns to scale

(D) Constant returns to scale

(E) Increasing returns to scale

**35.** Which of the following explains the difference between short-run and long-run costs?

(A) All costs are variable in the short run but not in the long run.

(B) All costs are fixed in the long run but not in the short run.

(C) All costs are variable in the long run but not in the short run.

(D) All costs are fixed in the short run but not in the long run.

(E) All costs are variable in the short run and fixed in the long run.